Run Pipeline

Description

Executes a targets pipeline from a specified script and saves progress visualization. Rejects profiles not authorized for the runner. Repeated failures are checked against the current execution progress even when their stored error message and last-success timestamp are unchanged. A thrown empty-branch dependency error retains expected-infeasible status only when current execution metadata prove a registered data root.

Usage

```
run_pipeline(
  sel_script,
  store_suffix = NULL,
  target_names = NULL,
  level_separation = 100,
  flag_validate_profile_selection = TRUE,
  vec_allowed_profile_roles = base::c("main", "smoke"),
  vec_allowed_profile_statuses = "active",
  plot_progress = TRUE,
  fresh_run = FALSE,
  prebuild_interpolation = FALSE,
  preparation_resource_profile = "shared",
  interpolation_workers_override = NULL,
  callr_function = callr::r
)
```

Arguments

`sel_script`	Path to the pipeline script to execute (relative to project root).
`store_suffix`	Optional character string appended as a sub-directory between the config-derived target store root and the pipeline name. When 'NULL' (default) the path is 'target_store/pipeline_name/' — identical to the original behaviour. When set, the path becomes 'target_store/store_suffix/pipeline_name/'. Useful when iterating over

	many spatial units that share one config but each need an isolated store (e.g. 'store_suffix = "eu_r01"').
`target_names`	Optional character vector of target names to build. 'NULL' builds the complete pipeline.
`level_separation`	Numeric value controlling the vertical separation between levels in the progress visualization network graph. Default is 100.
`flag_validate_profile_selection`	Logical controlling validation of the active profile role and status. Disable only for isolated tests that inject all configuration behavior.
`vec_allowed_profile_roles`	Profile roles authorized by this runner. Defaults to production roles 'c("main", "smoke")'.
`vec_allowed_profile_statuses`	Profile statuses authorized by this runner. Defaults to '"active"'.
`plot_progress`	Logical indicating whether to save a progress visualisation after the pipeline completes. Default is TRUE.
`fresh_run`	Logical indicating whether to destroy the existing target store before running the pipeline, forcing all targets to be re-computed from scratch. When 'TRUE', calls 'targets::tar_destroy(destroy = "all", store = ...)' prior to 'targets::tar_make()'. Default is 'FALSE'.
`prebuild_interpolation`	Logical indicating whether to prebuild 'data_community_interpolated' using parallel dynamic branches before running the complete pipeline. The worker count is read from 'data_processing$n_interpolation_workers' and resolved through 'preparation_resource_profile'. The default 'shared' profile caps workers at four, while 'dedicated' caps them at eight. Default is 'FALSE'.
`preparation_resource_profile`	Runtime resource profile passed to [resolve_preprocessing_worker_count()]. Defaults to 'shared'.
`interpolation_workers_override`	Optional explicit positive interpolation worker count.
`callr_function`	Function used by [targets::tar_make()] to launch the pipeline process. Defaults to [callr::r()]. Supply 'NULL' only for small graphs that are safe to execute in the current orchestration process.

Details

The function constructs pipeline-specific target store paths based on the script name and active configuration. When 'fresh_run = TRUE', the target store is destroyed with 'targets::tar_destroy(destroy = "all")' before execution so that every target is rebuilt from scratch. It uses 'targets::tar_make()' to execute the pipeline and then calls 'save_pipeline_progress_visualisation()' to generate a network visualization of the pipeline status. When 'prebuild_interpolation = TRUE', the function first executes only 'data_community_interpolated' with the 'crew_mori' preprocessing backend. Once that call returns, the complete pipeline is executed with [targets::tar_make()]. Thus downstream GPU-dependent model targets remain sequentially scheduled. A failed prebuild stops the complete run.

Value

No return value. Function is called for side effects: executes the targets pipeline and saves progress visualization to the documentation folder.

See Also

[save_pipeline_progress_visualisation()], [targets::tar_make()], [crew::crew_controller_local()]